

Data Collection

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[!IMPORTANT]

- The factory image already includes preset datasets required for training traffic sign recognition and lane detection models. This tutorial is provided for users who need to train their own YOLOv11 models and want to collect their own datasets
- If you want to experience the complete functionality directly, you can skip this section!!!

1. Course Content

1. Master the data collection methods required for training YOLOv11 object detection models. Training YOLOv11 object models requires first preparing datasets containing the objects to be detected

2. Traffic Sign Recognition Model Data Collection

2.1 Principle Explanation

- Record videos of targets to be detected under different lighting conditions and backgrounds through the camera, then obtain the original image dataset by capturing video frames at fixed intervals

2.2 Data Collection

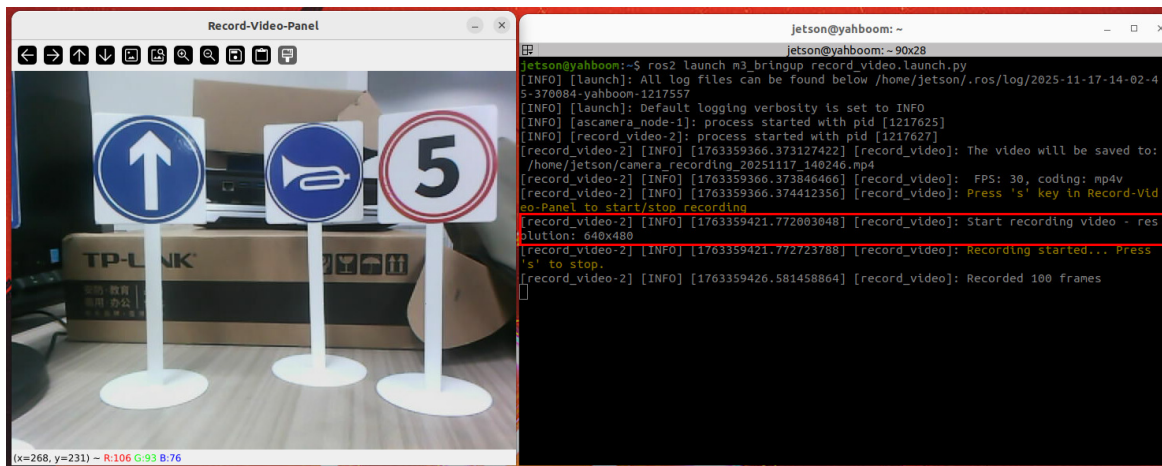
- Launch the data collection recording node

```
# Start the camera
ros2 launch ascamera hp60c.launch.py
# Record
ros2 run auto_drive record_video output_path:=~
```

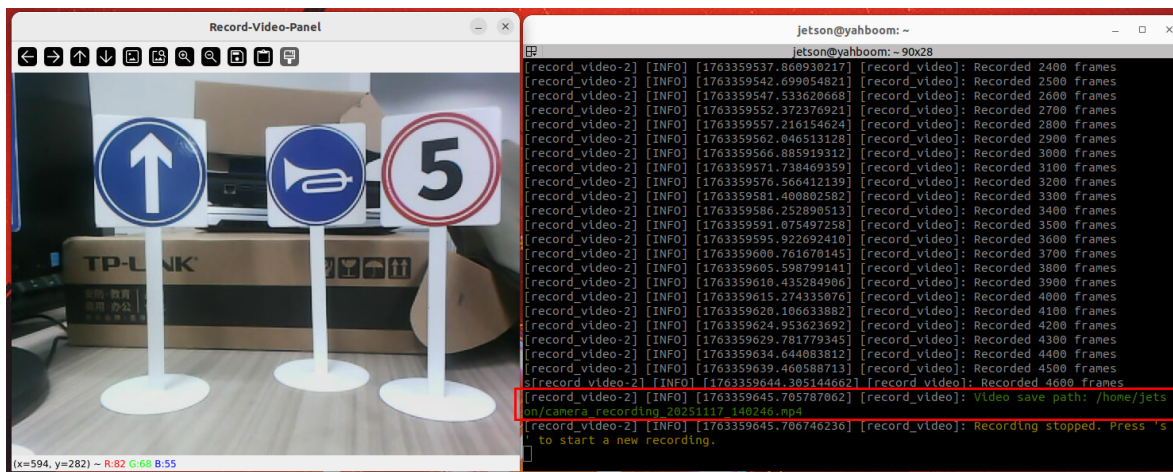
Optional Launch Parameters

- **output_path:** Path to save video files
 - **fps:** Frame rate of recorded video
-

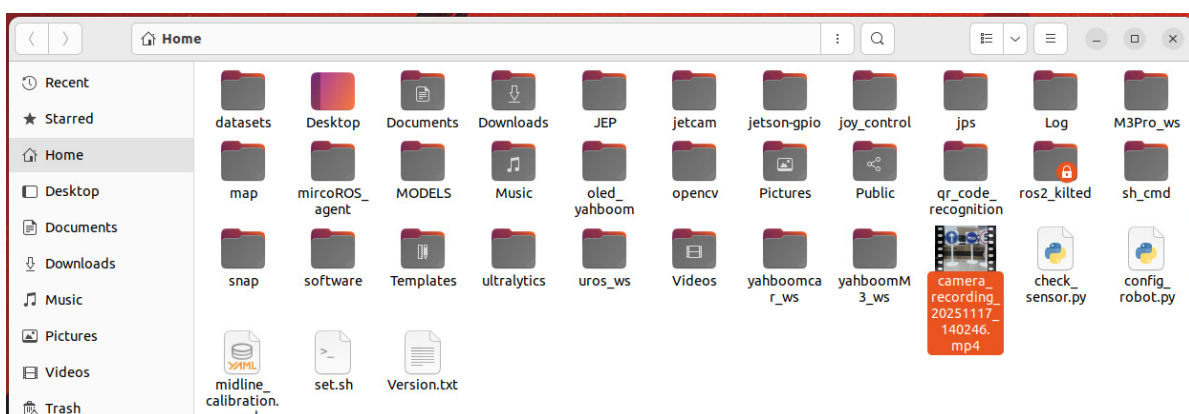
- After running the program, a video window will open. You can record video materials containing targets to be detected from different angles and under different lighting conditions by moving the chassis, which can enrich the diversity of the dataset.
- Place the traffic signs to be recognized within the view and start recording video



- Press Ctrl+C to end recording, and the terminal will print the path of the saved file



- If no video output path is specified, video files will be saved in the system user's home directory by default



- Split video frames into image sets. Place split_video.py from the "Model Training" folder in the source code summary to the root directory, then run the following command

```
cd ~
```

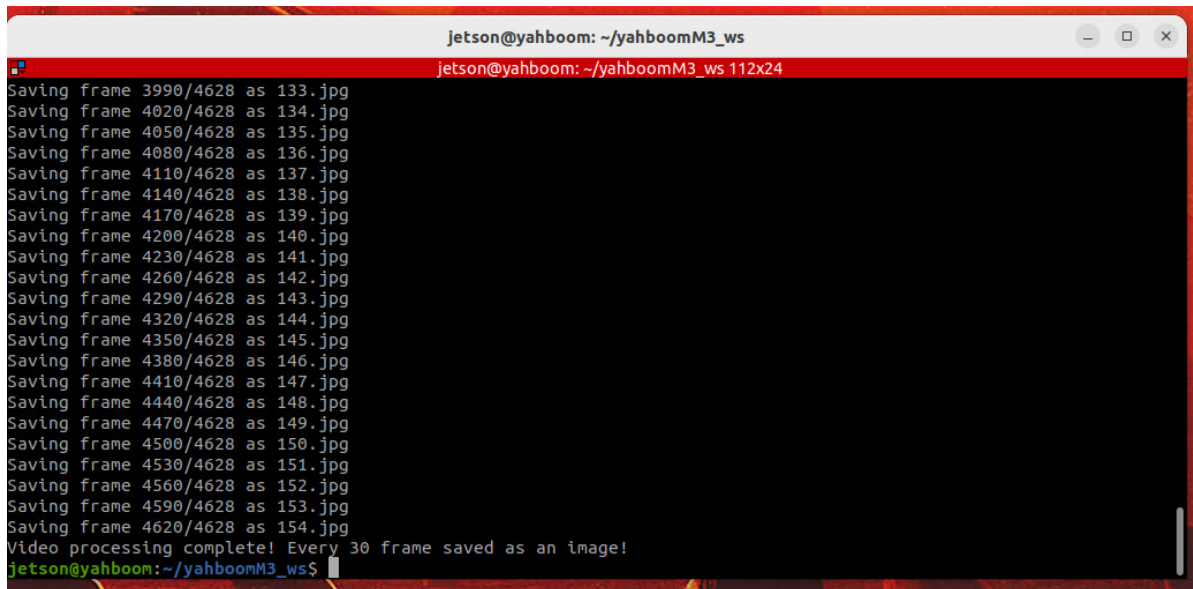
- Modify output_dir and video_file to your own image output path and video path

```
python3 split_video.py --output_dir ~/output_frame --video_file ~/camera_recording_20251117_140246.mp4
```

Parameter Description

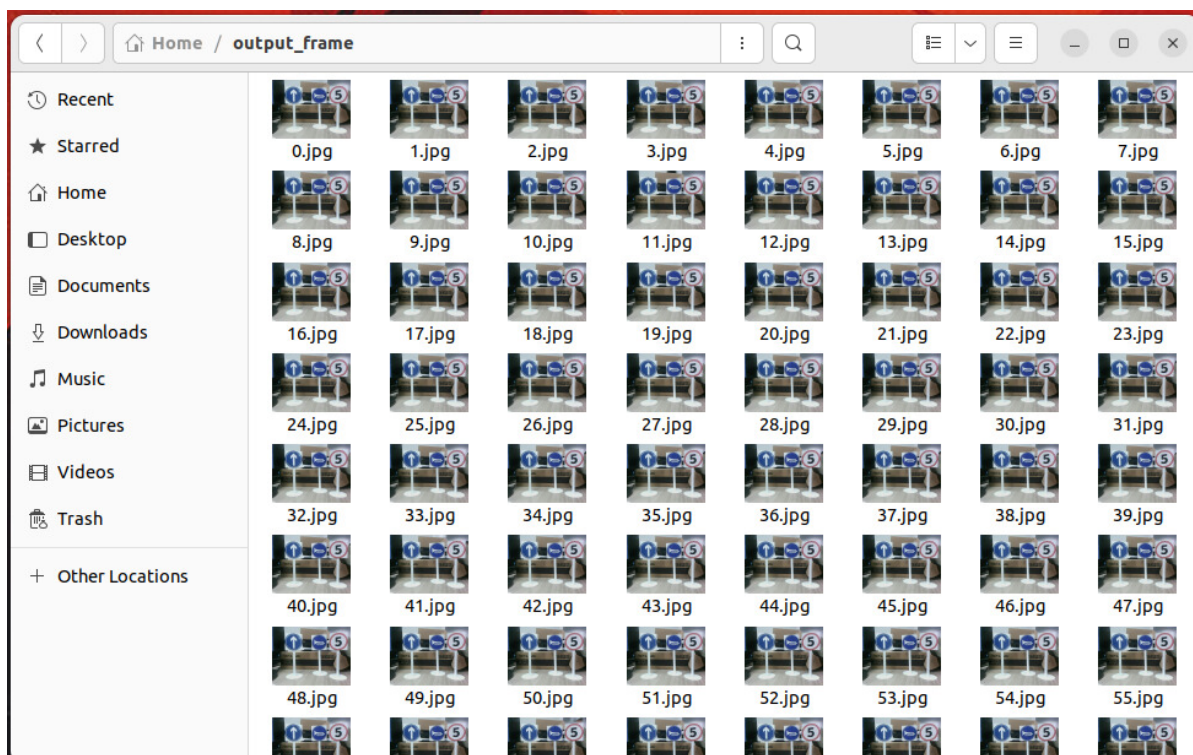
- **video_file:** Path of the video to be split
- **output_dir:** Path to save the split images
- **start_number:** Starting number for images
- **frame_interval:** Video frame interval, specifying how many frames to skip before capturing an image

- After splitting video frames is completed, the terminal will print prompt information



```
jetson@yahboom: ~/yahboomM3_ws
jetson@yahboom: ~/yahboomM3_ws 112x24
Saving frame 3990/4628 as 133.jpg
Saving frame 4020/4628 as 134.jpg
Saving frame 4050/4628 as 135.jpg
Saving frame 4080/4628 as 136.jpg
Saving frame 4110/4628 as 137.jpg
Saving frame 4140/4628 as 138.jpg
Saving frame 4170/4628 as 139.jpg
Saving frame 4200/4628 as 140.jpg
Saving frame 4230/4628 as 141.jpg
Saving frame 4260/4628 as 142.jpg
Saving frame 4290/4628 as 143.jpg
Saving frame 4320/4628 as 144.jpg
Saving frame 4350/4628 as 145.jpg
Saving frame 4380/4628 as 146.jpg
Saving frame 4410/4628 as 147.jpg
Saving frame 4440/4628 as 148.jpg
Saving frame 4470/4628 as 149.jpg
Saving frame 4500/4628 as 150.jpg
Saving frame 4530/4628 as 151.jpg
Saving frame 4560/4628 as 152.jpg
Saving frame 4590/4628 as 153.jpg
Saving frame 4620/4628 as 154.jpg
Video processing complete! Every 30 frame saved as an image!
jetson@yahboom: ~/yahboomM3_ws$
```

- The split images will be saved in the folder specified by output_dir



3. Lane Detection Model Dataset Collection

The lane detection model data collection process is the same as the traffic sign recognition model data collection process

3.1 Data Collection

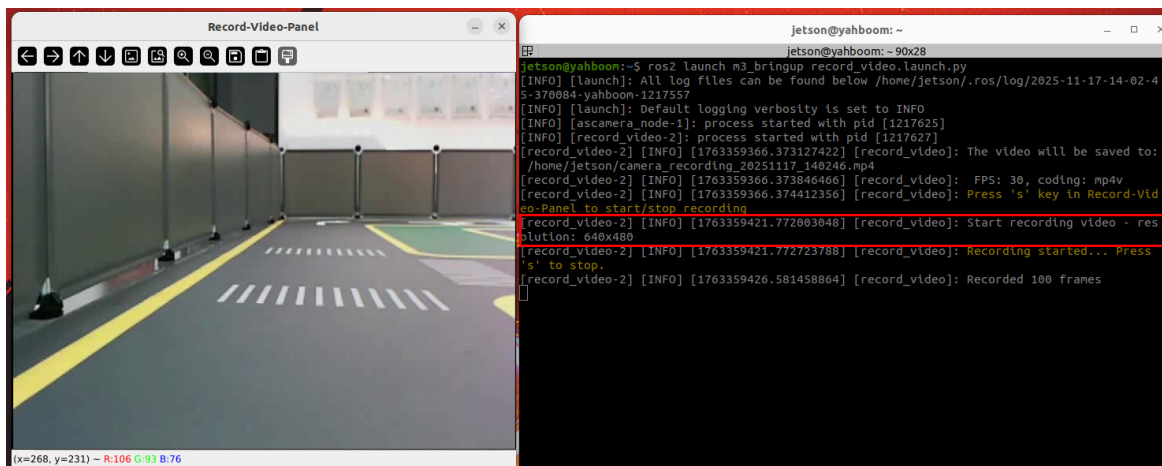
- Launch the data collection recording node

```
# Start the camera
ros2 launch ascamera hp60c.launch.py
# Record
ros2 run auto_drive record_video output_path:=~
```

Optional Launch Parameters

- **output_path:** Path to save video files
- **fps:** Frame rate of recorded video

- Place the mouse over the Record-Video-panel window and press the 's' key to start recording video



- The video saving and splitting methods are completely consistent with the previous ones, so they will not be repeated here.

4. Source Code Explanation

4.1 Recording Video from Video Topic

- The source code is located in record_video.py in the auto_drive package, source code path:

```
#orin nano \ rdx5
~/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/auto_drive/record_video.py
```

- The image callback function (image_callback method) is triggered every time an image message is received, responsible for:
Format conversion: Convert ROS image messages (msg) to OpenCV format (cv_image, bgr8 encoding, corresponding to RGB color space) through self.bridge.imgmsg_to_cv2.
Display image: Use cv2.imshow to pop up a window named Record-Video-Panel to display images in real-time.

Key control: Detect key presses through `cv2.waitKey(1)` (1ms delay, ensuring window refresh):

Press 's' key: Toggle recording state (start if not recording, stop if recording).

Recording logic: If `self.is_recording` is True, write the current frame to video through `self.out.write(cv_image)` and count (print log every 100 frames).

Exception handling: Catch errors in conversion or recording, prompt with red log.

```
def image_callback(self, msg):
    try:
        cv_image = self.bridge.imgmsg_to_cv2(msg, desired_encoding='bgr8')

        cv2.imshow("Record-Video-Panel", cv_image)
        key = cv2.waitKey(1) & 0xFF

        if key == ord('s'):
            if not self.is_recording:
                # Start recording
                self.start_recording(cv_image)
            else:
                # Stop recording
                self.stop_recording()

        # If recording is in progress, write the frame.
        if self.is_recording and self.out is not None:
            self.out.write(cv_image)
            self.frame_count += 1
            if self.frame_count % 100 == 0:
                self.get_logger().info(f"Recorded {self.frame_count} frames")

    except Exception as e:
        self.get_logger().info(Fore.RED+f"Error processing image: {str(e)}"+Fore.RESET)
```

- Recording control methods

start_recording method (start recording)

Initialize `cv2.VideoWriter`:

Get the width and height of the first frame (`first_frame.shape[:2]`, OpenCV format is (height, width)).

Parse encoding format with `cv2.VideoWriter_fourcc` (fourcc_code, e.g., mp4v corresponds to *'mp4v').

Concatenate video save path, create `VideoWriter` instance (parameters: path, encoding, frame rate, width and height).

Validate initialization: If `VideoWriter` failed to open (e.g., path error or unsupported encoding), print red error log.

Update status: `self.is_recording = True`, and print green log prompt for recording start.

stop_recording method (stop recording)

Update status: `self.is_recording = False`.

Release resources: If `self.out` exists (i.e., recording is in progress), call `self.out.release()` to close the video file, ensuring data is written completely.

Reset count: `self.frame_count = 0`, `self.out = None` (re-initialize for next recording).

Print logs: video save path (green) and re-recording prompt (yellow).

```

def start_recording(self, first_frame):
    """Start recording video"""
    if self.out is None:
        self.frame_height, self.frame_width = first_frame.shape[:2]
        fourcc = cv2.VideoWriter_fourcc(*self.fourcc_code)
        video_path = os.path.join(self.output_path, self.video_name)
        self.out = cv2.VideoWriter(
            video_path,
            fourcc,
            self.fps,
            (self.frame_width, self.frame_height)
        )

        if not self.out.isOpened():
            self.get_logger().error(Fore.RED+"Unable to open video editor.
Please check the encoding format and output path."+Fore.RESET)
            return

        self.get_logger().info(Fore.GREEN+f"Start recording video -
resolution: {self.frame_width}x{self.frame_height}"+Fore.RESET)

        self.is_recording = True
        self.get_logger().info(Fore.YELLOW+"Recording started... Press 's' to
stop."+Fore.RESET)

    def stop_recording(self):
        """Stop recording video"""
        self.is_recording = False
        if self.out is not None:
            self.out.release()
            self.get_logger().info(Fore.GREEN+f"Video save path:
{os.path.join(self.output_path, self.video_name)}"+Fore.RESET)

            self.out = None
            self.frame_count = 0

        self.get_logger().info(Fore.YELLOW+"Recording stopped. Press 's' to
start a new recording."+Fore.RESET)

```

4.2 Launch Program

- Used to start camera driver and the node for recording video from camera image topic in 4.1
- Source code is located in record_video.launch.py in the auto_drive package, source code path:

```

#orin nano \ rdx5
~/yahboomcar_ros2_ws/yahboomcar_ws/src/auto_drive/launch/record_video.launch.py

```

- Read camera type from environment variable CAMERA_TYPE:
If it's nuwa: Start hp60c.launch.py camera driver, subscribe to image topic /ascamera_hp60c/camera_publisher/rgb0/image.
- Node startup logic
First start the appropriate camera driver (bringup_camera_launch) to ensure image stream is published normally.
Then start the video recording node (record_video_node), passing parameters: save path,

frame rate, and corresponding camera's image topic.

The recording node (corresponding to the previous ImageToVideoNode) will subscribe to the specified image topic and execute recording function.

```
import os
from ament_index_python.packages import get_package_share_directory
from launch import LaunchDescription
from launch_ros.actions import Node
from launch.launch_description_sources import PythonLaunchDescriptionSource
from launch.actions import IncludeLaunchDescription, TimerAction
from launch.actions import DeclareLaunchArgument
from launch.substitutions import LaunchConfiguration

def generate_launch_description():
    CAMERA_TYPE = os.environ['CAMERA_TYPE'] #Get camera type
    m3_bringup_dir=os.path.join(get_package_share_directory('auto_drive'))
    if CAMERA_TYPE == 'nuwa':
        bringup_camera_launch = IncludeLaunchDescription(
            PythonLaunchDescriptionSource([
                os.path.join(get_package_share_directory('ascamera'), 'launch'),
                '/hp60c.launch.py'
            ])
        )

    elif CAMERA_TYPE == 'usb':
        bringup_camera_launch = IncludeLaunchDescription(
            PythonLaunchDescriptionSource([
                os.path.join(get_package_share_directory('usb_cam'), 'launch'),
                '/camera.launch.py'
            ])
        )

    #Startup parameters
    declared_arguments = []
    declared_arguments.append(
        DeclareLaunchArgument(
            "output_path",
            default_value="~",
            description="Video save path/视频保存路径 ",
        )
    )
    declared_arguments.append(
        DeclareLaunchArgument(
            "fps",
            default_value=30,
            description="Video frame rate/视频帧率 ",
        )
    )

    output_path=LaunchConfiguration('output_path')
    fps = LaunchConfiguration('fps')

    record_video_ndoe=Node(
        package='auto_drive',
```

```

        executable='record_video',
        name='record_video',
        output='screen',
        parameters=[
            {'output_path': output_path},
            {'fps': fps},
            {'image_topic': rgb_topic},
        ],
    )
    return LaunchDescription([
        *declared_arguments,      #Declare startup parameters
        bringup_camera_launch,    #Start the camera driver
        record_video_ndoe,        #Start recording video node
    ])

```

4.3 Video to Image Set Splitting Program

- Source code path:

~/Rosmaster/Sample/split_video.py

- Core process (main function)
 - Initialize configuration:
 - Read command line arguments to get video path, output folder and other configurations.
 - Automatically create output folder (if it doesn't exist).
 - Video processing:
 - Use cv2.VideoCapture to open video file, get total frames, frame rate and other information.
 - Loop through video frames, save 1 image every frame_interval frames (e.g., every 30 frames):
 - Image naming starts from start_number and increments sequentially (e.g., 0.jpg, 1.jpg).
 - Print progress in real-time (current frame / total frames + saved image name).

```

import cv2
import os
import argparse
def get_args():
    parser = argparse.ArgumentParser()
    parser.add_argument('--output_dir', default='engine', help='Output folder',
type=str)
    parser.add_argument('--frame_interval',default=30 ,help='Capture an image at
specified intervals from the video frames', type=int)
    parser.add_argument('--video_file', default='', help='Original video path',
type=str)
    parser.add_argument('--start_number', default=0, help='Image starting
number', type=int)
    return parser.parse_args()

if __name__ == "__main__":
    args = get_args()

```



```

# Path to the input video file
video_file =args.video_file
# Path to the output folder to save the frames
output_folder =args.output_dir
# Check if the output folder exists, if not, create it
if not os.path.exists(output_folder):
    os.makedirs(output_folder)
# Open the video file using OpenCV
cap = cv2.VideoCapture(video_file)
# Get the total number of frames in the video
total_frames = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
# Get the frames per second (fps) of the video
fps = cap.get(cv2.CAP_PROP_FPS)
# Frame counter to keep track of the current frame number
frame_number = 0
# Define the interval for extracting frames (every 15th frame)
frame_interval = args.frame_interval
# Initialize a new counter for saved frames (starting from 1)
start_number = args.start_number
while True:
    ret, frame = cap.read()
    # If the frame is not successfully read, exit the loop
    if not ret:
        break
    # Save every 15th frame
    if frame_number % frame_interval == 0:
        # Format the frame number starting from 1 (e.g., 1.png, 2.png)
        image_name = f'{start_number}.jpg'
        image_path = os.path.join(output_folder, image_name)
        # Save the current frame as a PNG image
        cv2.imwrite(image_path, frame)
        print(f'Saving frame {frame_number}/{total_frames} as {image_name}')
        # Increment the saved frame counter
        start_number += 1
    # Increment the frame counter for the video
    frame_number += 1

# Release the video capture object
cap.release()
# Print message when processing is complete
print(f"Video processing complete! Every {frame_interval} frame saved as an
image!")

```